

Southern California CSU DNP Consortium

California State University, Fullerton
California State University, Long Beach
California State University, Los Angeles

NON-VENTILATOR HOSPITAL-ACQUIRED PNEUMONIA PREVENTION
THROUGH ORAL CARE

A DOCTORAL PROJECT

Submitted in Partial Fulfillment of the Requirements

For the degree of

DOCTOR OF NURSING PRACTICE

By

Nora Buquid Catipon

Doctoral Project Committee Approval:

Raymund Gantioque, DNP, RNFA, ACNP-BC, Team Leader
Cinthya Sotelo, DNP, FNP-C, ENP-C, Team Member

May 2023

Author Note

Nora Buquid Catipon -  <https://orcid.org/0000-0003-0726-9056>

DOI 10.5281/zenodo.10548196

© 2023 Nora Buquid Catipon

ABSTRACT

Non-ventilator hospital-acquired pneumonia (NV-HAP) is a commonly identified classification of hospital-acquired pneumonia associated with high mortality, increased hospital length of stay, hospitalization costs, and readmission rates within 30 days. There is a lack of prevention strategies, surveillance, and reporting standards to track NV-HAP incidents. The literature provided evidence of the effectiveness of oral care in preventing NV-HAP. The purpose of this quality improvement (QI) project was to educate the nursing staff on a standardized, evidence-based oral care protocol to decrease the occurrence of NV-HAP. The project occurred over two months among acute stroke non-ventilated patients in a Neuro Intensive Care Unit. The Iowa model of evidence-based practice guided the planning and implementation of this QI project. The participants were a convenience sample of 46 registered nurses. Nurses' knowledge pre-and post-implementation of the oral care protocol was assessed. Aggregate data on NV-HAP occurrences were collected post-implementation. Chi-square test results showed an increase in nurses' knowledge of the NV-HAP diagnosis post-implementation (78.8%) from pre-implementation (60%) of an oral care protocol. There was an increase in nurses' understanding of the importance of using an oral care protocol to prevent NV-HAP pre-implementation (42.5%) and post-implementation (46.9%). No NV-HAP occurred post-implementation, with two incidents pre-implementation. This QI project showed that using an oral care protocol can decrease the occurrence of NV-HAP. Nurses are empowered to implement routine oral care protocols as a preventive strategy for healthcare-associated infections, leading to improved clinical outcomes and a decrease in the economic burden placed on hospitals.

Keywords: oral care, oral hygiene, oral hygiene care, nursing care, healthcare-associated, pneumonia, complications, economics, nursing, pneumonia organization and administration, prevention and control, pneumonia therapy

TABLE OF CONTENTS

ABSTRACT.....	iii
ACKNOWLEDGEMENTS.....	vii
BACKGROUND	1
Oral Care as a Modifiable Risk Factor	2
Problem Statement.....	3
Purpose Statement.....	3
Project Objectives	4
Supporting Framework: Iowa Model of Evidence-Based Practice.....	4
IOWA Model Steps and Decisions	5
Application of the Iowa Model	7
REVIEW OF LITERATURE	10
Hospital-Acquired Pneumonia.....	11
A Call to Action to Prevent NV-HAP.....	11
Risk Factors for the Development of NV-HAP	12
Patient and Nursing Perspective of Oral Cares	12
Oral Care as a Risk Mitigation Strategy for NV-HAP Prevention	13
Additional Risk Mitigation Strategies for NV-HAP Prevention	16
METHODS	19
Design	19
NV-HAP Intervention Audit Tool	19
Setting	20
Participants.....	21
Institutional Letter of Support and Institutional Review Board Approval	21
Planning the QI Program	22
Develop an Implementation Plan.....	22
Phase 1-Create Awareness and Interest	23
Phase 2-Build Knowledge and Commitment.....	23
Phase 3-Promote Action and Adoption.....	23
Phase 4-Pursue Integration and Sustained Use of the Oral Care Protocol	25
Measures	25
Data Collection and Management.....	25

RESULTS	27
Data Analysis	27
DISCUSSION	29
Limitations	30
Recommendations	31
Conclusions	32
REFERENCES	34
APPENDIX A Iowa Model	41
APPENDIX B Oral Care Protocol	43
APPENDIX C: Oral Care Protocol Audit Tool	44
APPENDIX D: Institutional Review Board Letter from CSLA	45
APPENDIX E: Institutional Review Board Letter from Project Hospital	46
APPENDIX F: Letter for Approval to Conduct at the Project Hospital	47
APPENDIX G: Implementation Phases of the Iowa Model	48
APPENDIX H: Pre-Implementation Survey	49
APPENDIX I: Post-Implementation Survey	52
APPENDIX J: Oral Care Patient Education Handout	55
APPENDIX K: Oral Care Patient Education Handout	56
APPENDIX L List of ICD 10 NV-HAP Codes	57
APPENDIX M: Nurses' Knowledge of the Importance of an Oral Care Protocol to Prevent Non-Ventilator Hospital Acquired Pneumonia (NV-HAP)	58
APPENDIX N: Nurses' Survey Response on the Duration of Oral Care	59
APPENDIX O: Nurses' Survey Response on the Prioritization of Oral Care	60

ACKNOWLEDGEMENTS

I humbly thank my team leader Dr. Raymund Gantioque for his guidance and support through each stage of my DNP scholarly writing project. I also would like to thank my team member Dr. Cinthya Sotelo for her guidance and for letting my oral defense be an enjoyable experience. My nurse leaders at my institution, Darlene Scafiddi, Kenny Cantiller, and Lolla Mitchell, were instrumental in the execution and completion of my DNP clinical project despite the staffing challenges undertaken during the winter months. For this, I am extremely grateful. I also extend my warmest gratitude to Dr. Elizabeth Winokur for her sincere guidance during my IRB application process approval and for defining the path to obtain meaningful data collection from the project participants. I want to express my heartfelt gratitude to my writing coach Dr. Gina Armendariz, who rendered guidance throughout the various stages of my DNP writing project which helped me enhance my writing skills.

I want to give my warmest thanks to my colleague and classmate, Maybetter Forte who inspired me to persevere and overcome fear. She encouraged me to start our DNP journey together as a long race, followed by many short races, one after the other and to finish strong. I will never look back to regretting this journey we embarked on together despite the sleepless nights and difficulties in balancing full-time school and work.

My joy and love know no bounds in expressing my jovial gratitude to my husband, daughter, sister, brothers, and other family members for their continuous support and understanding which carried me through my DNP coursework. Special thanks to my sister, my friend who helped me carry the torch to the end of my DNP journey.

Background

Non-ventilator hospital-acquired pneumonia (NV-HAP) is one of the most common hospital-acquired infections and is associated with substantial negative impact on clinical outcomes, including increased morbidity and mortality and economic burdens (Guiliano et al., 2021). It is estimated that 1 in every 100 hospitalized patients will be affected by NV-HAP, but currently, there are no surveillance methods, tracking, reporting system, or preventive strategies to prevent hospital-acquired infection (Munro et al., 2021). Experts from the National Organization to Prevent Hospital-Acquired Pneumonia (NOHAP) and the Joint Commission are calling an action to address NV-HAP at a national level to support strategic interventions to prevent NV-HAP, educate the public and practitioners, and develop NVHAP surveillance programs and prevention guidelines (Munro et al., 2021).

Hospital-acquired pneumonia, the most common healthcare-associated infection in the United States, is associated with increased mortality and morbidity (Fine, 2020). It usually develops 48 hours after hospitalization and includes two distinctive categories of disease: non-ventilator hospital-acquired pneumonia (NV-HAP) and ventilator-associated pneumonia (VAP). Considerable research on VAP prevention has been the focus leading to initiation of evidence-based preventive care bundles. The efforts and initiatives led to significant decline in VAP rates, improved patient outcomes, and decreased healthcare costs related to VAP (Guiliano et al., 2018). According to Davis and Finley (2012), NV-HAP was identified to be more common than VAP, occurring on every type of hospital unit with a higher incidence of mortality of 18% and higher costs than VAP. NV-HAP crude mortality rate is 15%-30%, extends hospital length of stay by up to 15 days, requires intensive care unit (ICU) admission in up to 46% of non-ICU cases, increases antibiotic utilization, and is associated with readmission within 30 days in up to

20% of survivors (Munro et al., 2021). Scant data exist on the economic impact of NVHAP alone, but given that NV-HAP accounts for 60% of hospital- acquired pneumonia (HAP), the impact is likely substantial (Magill et al., 2018; Munro et al., 2021). It is recognized that knowledge gaps of NVHAP pathogenesis, prevalence, complication rates, clinical outcomes, lack of nursing resources, and increased healthcare demands are creating challenges in implementing system-wide NV-HAP prevention strategies. There is an emerging body of evidence on the effectiveness of oral care in preventing NV-HAP. Quinn et al. (2014), reported a 37% decrease in NV-HAP and cost avoidance of an estimated \$1.6 million over 12 months on implementation of a standardized oral care program. By far, oral care is the best-studied NVHAP prevention intervention. However, findings in the literature reported missed oral care opportunities for hospitalized patients, lack of standardized oral care protocols, and surveillance systems to track NV-HAP incidents.

At an acute care facility in Southern California, there were approximately 75 patients that developed NV-HAP for the last two years in the Neuro ICU Neuro Intensive Care Unit (ICU). Currently, there exist no preventive strategies, such as a standardized oral care protocol utilized in nursing practice for non-ventilated patients to prevent NV-HAP. The purpose of this Doctor of Nursing Practice (DNP) project is to implement a standardized oral care protocol to decrease the occurrence of NV-HAP incidents in a 12-bed Neuro ICU department.

Oral Care as a Modifiable Risk Factor

Aspiration is believed to be the primary route for acquisition of NV-HAP. Gram-negative bacteria are responsible for most bacterial cases of NV-HAP, with rates ranging from 50%-80%. Up to 45% of adults silently experience micro-aspiration and impaired clearance of oral and oropharyngeal secretions that facilitate the inoculation of potential pathogens into the lungs

(Fine, 2020). In the absence of regular oral care, a biofilm quickly develops and is colonized by harmful bacteria from the mouth and the lungs, which double every two to three hours. Bacteria can also be transmitted through self-inoculation and hand-to-face transmission. (Fine, 2020). Influenza and RSV infections contribute substantially to the morbidity and mortality associated with NV-HAP (Nabeya et al., 2017; Vanhems et al., 2016). Munro and Baker (2018) found that mechanical removal of oral biofilm through simple tooth brushing reduced the relative risk of pneumonia. In addition, providing consistent oral care two to four times a day may decrease the risk of NV-HAP by 40-60%. Prevention of NV-HAP by providing oral care is the most common modifiable risk factor and has the potential to improve quality patient care and patient safety and potentially improve customer satisfaction.

Problem Statement

The problem this project addresses is poor oral health which is associated with a high prevalence of NV-HAP rates in an acute hospital setting in Southern California. NV-HAP is one of the leading causes of hospital-acquired infections. Despite public health concerns, increased mortality, and morbidity associated with NV-HAP, there is a lack of prevention strategies, surveillance, and reporting to track this condition.

Purpose Statement

The purposes of this DNP project are to educate the nursing staff on a standardized, evidence-based oral care protocol and decrease the occurrence of NV-HAP by 25% over two months period among acute stroke non-ventilated patients in a twelve-bed Neuro Intensive Care Unit of an acute care community hospital located in Los Angeles County, California.

Project Objectives

The objectives of this doctoral project are:

- a. Improve the frequency of oral assessment and documentation of oral care provided every shift after the implementation of the standardized oral care protocol
- b. Develop and implement a standardized, evidence-based oral care protocol and guidelines of care.
- c. Identify the NV-HAP rates prior to and post-implementation of a standardized protocol in the Neuro Intensive Care Unit
- d. Collaborate with the Information Technology department to modify the Electronic Medical Record to support nursing documentation of the oral care protocol and develop reports.

Supporting Framework- Iowa Model of Evidence-Based Practice

A theoretical framework provides a structured approach for nurses and clinicians to identify clinical problems, develop and implement strategies, and evaluate quality improvement projects using evidence-based practice (Tucker et al., 2021). The Iowa Model of Research-Based Practice to Promote Quality Care was developed to guide clinicians in evaluating and infusing research findings into patient care (Buckwalter et al., 2017; Titler et al., 2001). It was revised in 2015 based on researcher feedback and knowledge expansion of research strategies (Titler et al., 2001). The Iowa Model's key attributes are based on Roger's innovation theory. These include alignment with organization priorities, team-based, step-by-step processes, critical decision points, feedback loops, phased implementation, specific implementation strategies, evaluation, and dissemination (Tucker et al., 2021). Use of the Iowa Evidence-Based Practice (EBP) theoretical framework enables the implementation of EBP, synthesis of high-quality research

literature in healthcare, and the development of clinical practice guidelines (Grove & Gray, 2019). According to Zhao et al. (2016), the Iowa Model is easy to implement for all nurses, from novice to expert, because of its simple, clear, concise, and application-oriented approach. Since its origin in 1994, Iowa Model has been continually referenced in nursing journal articles and extensively used in clinical research programs in a variety of settings to guide decision-making and implementation for practice change (Zhao et al., 2016).

IOWA Model Steps and Decisions

The first step in the Iowa Model begins with identifying triggering problems that evolve from clinical issues, data, or new evidence. (Cullen et al., 2018; Titler et al., 2001). Triggering issues can be generated when considering organization, state, or national initiatives such as patient outcomes or financial reimbursement. Once a triggering event is identified, it is essential to engage potential stakeholders to collaborate and clarify priority issues and opportunities to develop a clear, specific purpose statement.

The second step in the Iowa model is the creation of a problem statement using the PICOT format (P = patient/problem/population, I = intervention, C = comparison, O = outcome, T= time). There are three important decision points embodied in the Iowa Model algorithm to help guide the clinician. The first decision point is a closed question asking if the problem is a priority for the organization. If the topic is not a priority for the organization, a feedback loop recommends that the researcher considers other clinical issues. If the project is feasible to implement and a priority to the organization, the next step is the formation of the EBP team.

The third step is the development of an interdisciplinary team composed of invested stakeholders committed to improving patient care and addressing the clinical issue (Titler et al., 2001). Once the team is created, the fourth step is to conduct an extensive literature search of

evidence-based guidelines, systematic research reviews, meta-analyses, and clinical studies. The interdisciplinary team will assemble the literature, critically appraise the evidence, and synthesize a body of evidence (Cullen et al., 2018).

The second decision point is to determine if there is sufficient evidence to guide practice. The synthesis of the literature will influence the interdisciplinary team's decisions as they move forward with the project. The fifth step is to design a pilot test of interventions for change. The purpose of the pilot is to determine whether the implementation plan will result in the adoption of the practice change and to assess readiness to roll out the interventions to other patient care areas (Cullen et al., 2018). The pilot process entails the following steps: (a) engage patients and verify preferences, (b) consider resources, constraints, and approval, (c) develop localized protocol, (d) create an evaluation plan, (e) collect baseline data, (f) develop an implementation plan, (g) prepare clinicians and materials, (h) promote adoption, and (i) collect and report post-pilot data (Titler et al., 2001).

The final decision point in the Iowa Model is to determine if the change is appropriate for adoption into practice. According to Titler et al. (2001), if the outcomes of the piloting of practice change are not achieved as expected, the team should consider alternatives such as evaluation of new knowledge from recent research. After the decision is made to fully integrate a practice change, additional work is needed to move from adoption to sustained use of the evidence-based practice protocol (Cullen et al., 2018). When the interdisciplinary team has adapted the interventions for practice, the sixth step is to integrate and sustain the practice change (Figure 2). Key elements to sustain change include identifying and engaging key personnel, hardwiring change into the system, monitoring key indicators of quality improvement, and reinforcement as needed through ongoing evaluation, education, and training (Cullen et al.,

2018). This phase of implementation continues to require local leadership to strengthen practice improvements and continued participation of champions and senior leaders to provide support of the EBP standard of care and process.

The seventh and final step in the Iowa Model is to disseminate results. Dissemination should occur internally within the organization through newsletters, shared governance, and quality improvement committee meetings. External dissemination may be accomplished through poster presentations and presentations at national conferences as well as peer-reviewed publications (Cullen et al., 2018).

Application of the Iowa Model

Application of the Iowa Model in this DNP project began first with the pre-assessment of a triggering issue. In this DNP project, the triggered clinical problem is lack of an oral care protocol to prevent NV-HAP. The purpose of quality improvement projects is to find a solution to a problem or compare new knowledge to current practice (Cullen et al., 2018). The Joint Commission has recommended that organizations implement strategies to prevent NV-HAP (The Joint Commission, 2021). The practice change is aligned with the organization's top priority to prevent hospital-acquired infection by providing high-quality, cost-effective, and safe care. The next step is the statement of a question using the PICOT format (see Appendix A, Figure 1).

After obtaining organizational support and an affirmative response to the first decision point, the next step was the formation of an interdisciplinary team. The interdisciplinary team consisted of the advanced practice nurse (APN), nurse champion, physician champion, and infection preventionist. The interdisciplinary team assembled, appraised, and synthesized the best available evidence related to the implementation of an oral care protocol in the prevention of

NV-HAP. Once the critical appraisal of literature was completed, the team determined that there was sufficient evidence to proceed with a pilot intervention.

As the interdisciplinary team decided to move forward based on the synthesized body of evidence, the next step was to design a plan to implement and pilot the practice change. An organization letter of support and IRB approval letter from the organization and the school were obtained before implementing the pilot trial. Pre-implementation baseline data of NV-HAP rates for three months before the implementation of the pilot project was collected by the project author. Implementation strategies included the development of a written standardized oral care protocol, creating an evaluation plan, and nursing staff education using checklists, posters, pocket guides, communication reminders, and information technology support. This DNP pilot project was projected to start at the end of the third quarter 2022.

Post-implementation data was collected, evaluated, and reported after two months. Barriers were analyzed, and individualized feedback was reported to the interdisciplinary team. The third and final decision point is to evaluate if the practice change is appropriate for adoption in practice. Based on the pilot process and patient outcomes data, a clinical decision will be made.

The interdisciplinary team must ensure the change is feasible and will result in improved outcomes before full-scale implementation (Brown, 2014; Zhao et al., 2016). If the pilot is successful, the practice change can be extended to other adult nursing departments. Key steps to move from adoption to sustained use of the oral care protocol after the pilot include the identification and engagement of nursing unit RN champions, the creation of a monthly dashboard report to track and trend data, hardwiring the practice change as the routine workflow, continued evaluation and monitoring of performance improvement indicators, and enhanced

education as needed. Implementing a routine oral care protocol to prevent NV-HAP hardwires the behavior change as the new norm. Change agents are pivotal in promoting sustainability and commitment to practice change. Some activities to sustain change include audits and feedback, celebration of local unit progress, individualizing data feedback, sharing protocol revisions with clinicians based on feedback, and updating practice reminders (Cullen et al., 2018). Additionally, building organizational support through senior leadership is accomplished by reporting to senior leaders, reporting at quality improvement program committees, revision of policy and oral care protocols, and implementing educational programs.

The final step in the Iowa model algorithm is the dissemination of results internally through the organization's unit-based and house-wide shared governance committee meetings. External dissemination of results from EBP projects will be accomplished through a poster presentation at local and national conferences and submission to pertinent peer-reviewed nursing journals.

Review of Literature

A review of the literature was conducted for this DNP quality improvement project. The following electronic databases were searched for publications relevant to the oral care and prevention of NV-HAP: CINAHL, PubMed, EBSCO host, and Google Scholar. MeSH terms were also used and included oral care, oral hygiene, oral hygiene care, nursing care, healthcare-associated (HCA), and pneumonia. Further search terms included HCA sub-headings, including complications, economics, nursing, pneumonia organization and administration, prevention and control, and pneumonia therapy. Articles were selected if they were peer-reviewed, written in English, and published between 2017-2022. Articles were selected from the last five years since this timeframe represents the most recent research. Reference lists of retrieved articles were also hand-searched to identify pertinent publications. Two search phrases were used: “oral care” and “non-ventilator hospital-acquired pneumonia.” A comprehensive literature search yielded 237 articles, 223 excluded as duplicates, interventions that did not test oral care, peer-reviewed, systematic reviews, posters, and/or editorials. This left 14 articles for the project. However, two articles from the 14 were excluded due to termination of the study because of the COVID-19 pandemic, and the other was excluded because it is currently an active study that has not yet been completed. After excluding the two articles, were 12 articles left for inclusion in the literature review. The 12 articles were synthesized and critiqued using Melnyk and Fineout-Overholt’s hierarchy of evidence (Grove & Gray, 2019). The next section will discuss the significance of NV-HAP, risk factors for developing NV-HAP, patient and nursing perspectives of oral care, and other risk mitigation strategies for NV-HAP prevention.

Hospital-Acquired Pneumonia

Hospital-acquired pneumonia (HAP), also called nosocomial pneumonia, is among the most common hospital-acquired infections [HAI] (Baker & Quinn, 2018). HAP occurs at a rate of up to 25 cases per 1,000 admissions (Baker & Quinn, 2018). It is named one of the top 10 public health concerns for patient safety by the Centers for Disease Control and Prevention [CDC] (Baker & Quinn, 2018; CDC, 2016; Magill et al., 2018). HAP is defined as an infection of the lower respiratory tract that does not exist at the time of admission and does not have an incubation period of infection but occurs 48 hours after admission (Baker & Quinn, 2018). HAP includes 2 distinct categories: non-ventilator HAP (NV-HAP) and VAP. NV-HAP and VAP are associated with significant clinical and financial burdens, such as prolonged hospital length of stay (LOS), higher overall healthcare costs, and increased morbidity and mortality rates (Guiliano et al., 2018; Saied et al., 2019).

According to Guiliano et al. (2018), NV-HAP is a serious patient safety concern in acute care, with reported incidence ranges between 3.63 per 1,000 patient days. An NV-HAP estimated acute care cost for treatment is between \$28,000 and \$40,000 per hospitalization (Guiliano et al., 2018). In a large cohort of patients, ICU NV-HAP and VAP were associated with 82% and 38% of the risk of 30-day mortality, respectively (Saied et al., 2019, Tang et al., 2020). Micek et al. (2016), and Baker and Quinn (2018), found that patients with NV-HAP had a median hospital LOS of 15 days compared to 4.4 days for patients who did not develop NV-HAP.

A Call to Action to Prevent NV-HAP

In 2020, a group of U.S. healthcare leaders from the National Organization to Prevent Hospital-Acquired Pneumonia (NOHAP) issued a national call to action to address NV-HAP awareness and prevention (Munro et al., 2021). The recommended strategies include patient,

healthcare professional, and student education on preventative measures; healthcare systems and insurers to support NV-HAP prevention; and development of new strategies for NV-HAP surveillance, tracking, and reporting of NV-HAP rates (Munro et al., 2021). Additionally, the CDC recommends focusing on modifiable risk factors to prevent HAIs (CDC, 2016; Tesoro et al., 2018). Oral care is one of the most common and inexpensive strategies used in various acute hospitals and long-term care settings to prevent NV-HAP, yet studies have shown variations in practices.

Risk Factors for the Development of NV-HAP

Some of the risk factors for the development of NV-HAP include older age, partial or complete immobility, depressed consciousness, sedation, paralysis, malnutrition, chronic illnesses such as renal failure, chronic obstructive disease, multiple co-morbidities, male sex, length of hospital stay, dysphagia, enteral feedings, and poor hygiene associated with dental plaque build-up (Klompas et al., 2022; Lukasewicz Ferreira et al., 2022; Mitchell et al., 2019; Talley et al., 2016).

Previous reviews indicated that strategies targeted at risk factors might decrease the risk of NV-HAP, including oral care, hand hygiene, early mobility, evaluation and management of post-stroke patients, dysphagia screening, and management. Yuan et al. (2020) found that male stroke patients were more likely to benefit from oral care due to poor oral hygiene associated with higher smoking rates and reduced respiratory resistance to infection.

Patient and Nursing Perspectives of Oral Care

Two qualitative studies provided perceptions of patients, nurses, and ancillary stroke clinicians on oral care for patients (Coker et al., 2020; Ferguson et al., 2020). The themes identified were knowledge gaps, staff barriers due to limited time and resources, and lack of

standardized oral care protocol that led to inconsistencies in oral care practices among nurses and ancillary stroke clinicians. These themes led to inconsistencies in oral care practices provided by nurses and ancillary stroke clinicians. The study revealed that patients primarily felt that oral health deterioration occurred during hospitalization. The patients valued oral care and were open to nurses assisting them with their oral care needs. (Coker et al., 2020; Ferguson et al., 2020).

Nurses must ensure that oral hygiene is provided.

Oral Care as a Risk Mitigation Strategy for NV-HAP Prevention

Oral care is a fundamental nursing procedure for hospitalized patients. It is the best-studied NV-HAP prevention strategy. The primary goal of oral care is to promote oral hygiene, decrease microbial colonization in the oropharynx and dental plaque, and reduce aspiration of contaminated saliva. (Feider et al, 2010; Khasanah et al., 2019). Poor oral care practices in acute care can lead to deterioration of oral health and the risk of developing pneumonia, which can impact a patient's overall well-being and quality of life (Quinn et al., 2014). Bacteria found in dental plaques may serve as a reservoir for respiratory pathogens in hospitalized patients combined with previously mentioned risk factors that contribute to developing NV-HAP. (Lukasewicz Ferreira et al., 2022; Talley et al., 2016). Moreover, aspiration of oropharyngeal secretions containing pathogenic bacteria from the upper airway into the lower respiratory tract is one of the most important mechanisms for NVHAP. Further studies have shown that improving oral hygiene effectively reduces patients at risk for NV-HAP.

Patients in the ICU often cannot effectively remove bacteria from their mouths due to illness, older age, or dysfunction (Tang et al., 2020). Several studies have shown that oral care could significantly reduce the incidence of VAP; however, this finding is not seen in hospitalized patients without ventilatory support.

Quality improvement (QI) studies have shown that prevention of NVHAP involves consistent oral care hygiene (Chik & Wynne et al., 2020; Munro & Baker, 2018; Lacerna et al., 2020; Pritts et al., 2020; Warren et al., 2019). Chik & Wynne (2020) and Pritts et al. (2020), showed that high fidelity to the oral care intervention reduced waste of time, improved compliance with oral care, and decreased NV-HAP rates. The availability of oral care products at the time of care led to improved oral hygiene. QI studies of nurse-led initiatives used collaboration with multidisciplinary team. The studies showed that implementing multidisciplinary QI teams using comprehensive methodologies and fidelity to oral care practices improved NV-HAP rates. The studies had limitations due to their small sample sizes, which can lead to a Type II error. However, the QI studies were of moderate to high-quality evidence.

Additionally, other studies have found similar barriers in performing and documenting oral care, such as limited time with competing demands, communication errors, staff knowledge gaps due to lack of education and training, and low prioritization compared to other nursing tasks (Jenson et al., 2018, Satheeshkumar et al., 2020; Tesoro et al., 2018). Furthermore, staff perceived oral care as a comfort measure rather than a nursing procedure to prevent HAIs. Gaps identified in the studies included a lack of consistent, standardized oral care policies and written protocols, inconsistent brushing frequencies, appropriate and adequate oral care supplies, documentation systems in the electronic records, and training of nurses and health care assistants.

A systematic review and meta-analysis of randomized clinical trials (RCTs) and control trials suggested that oral care reduces the incidence of NV-HAP primarily when a dental care professional and dedicated nursing assistant were involved in caring for patients in nursing homes. The study was conducted in acute and long-term care settings leading to high

heterogeneity among participants (Satheeshkumar et al., 2020). Not only was the study population and setting different, but there were inconsistencies in delivery of oral hygiene, leading to infidelity in intervention implementation. This could potentially influence the findings. Most RCTs were conducted in nursing homes, limiting the findings' external validity. This demonstrates the need to conduct RCTs in acute settings for oral care.

Similar findings were found in a systematic review of RCT, cohort, and quasi-experimental studies. Improvements in NV-HAP were seen when oral care was provided by a dentist or a dental hygienist in three nursing homes. The studies used for this systematic review were retrieved from one database (Medline). This may have resulted in missed articles, which could have been found had the researchers explored several databases. Despite the heterogeneity found in oral care interventions (different antiseptics) used in the studies, there was a reduction in NV-HAPs (Mitchell et al., 2019).

The group of studies included in the systematic review and meta-analysis were sufficiently valid in evaluating the effectiveness of various strategies to reduce NV-HAP (Mitchell et al., 2019; Satheeshkumar et al., 2020). Oral care interventions were used in hospitals and nursing homes, resulting in similar positive findings associated with the reduction of NV-HAP (Mitchell et al., 2019; Satheeshkumar et al., 2020). The study findings were difficult to generalize since there was high heterogeneity in the interventions used and a probable wide age range in the patient population, along with a small size across a variety of healthcare settings, made the generalizability of findings difficult. Due to limited generalizability, it may be difficult to translate these findings into the project setting.

In Giuliano et al. (2021), the RCT study was conducted in an acute care setting of a single medical center. The study aimed to determine the effectiveness of a universal standardized

oral care protocol in preventing hospital-acquired pneumonia (HAP). The study findings revealed the odds of developing NV-HAP were 7.1 times in the medical control versus intervention units. Frequency of oral care increased with an 85% reduction in the NV-HAP incidence rate (Giuliano et al., 2021).

The reported findings of the systematic review and meta-analysis showed that RCTs were insufficient to draw substantial conclusions due to lack of standardized oral care practices conducted in acute hospital settings and long-term care settings. Therefore, robust RCTs with a larger sample size using a standardized oral care protocol are needed for hospitalized patients to evaluate the effectiveness of the oral care protocol in preventing NV-HAP (Mitchell et al., 2019; Satheeshkumar et al., 2020). Additionally, study findings will provide guidance in the development of standardized oral care practice guidelines to mitigate risk of NV-HAP.

Additional Risk Mitigation Strategies for NV-HAP Prevention

There is some evidence to support the effectiveness of dysphagia and physical activity in lowering the risk of NV-HAP. Two non-randomized studies were conducted in a hospital using dysphagia screening as the primary method of NV-HAP prevention for stroke patients that indicated a positive direction in reducing pneumonia. Two other RCT studies using a small sample size performed physical activity in hospitals, which significantly reduced pneumonia in non-ventilated patients. In a quality improvement study, multiple interventions, including increased activity, respiratory care, oral care, and education, were associated with decreased rates of NV-HAP and decreased mortality rates (Lacerna et al., 2020).

Hospitalized patients may suffer progressive declines in their mobility leading to muscle atrophy was reported to occur at a rate of 3%-11% per day. (Munro et al., 2021. Physical activity is a second promising strategy to prevent NVHAP. In Mitchell et al. (2019), an RCT was

conducted in acute stroke patients using a turn-mobility program. The procedure used routine patient turning every two hours with combined active and passive range of motion of the four limbs every four hours, which led to reduced incidence of NV-HAPs. In Boden et al. (2018), the interventions reported to reduce NV-HAP were pre-operative patient education, early ambulation, and self-directed breathing exercises. Mobility programs and dedicated healthcare teams, including physical therapists and nurses, can help maintain patient conditioning and prevent NV-HAPS.

Early dysphagia screening in high risk could help prevent HAP. Several studies have shown that early dysphagia screening within 24 hours of admission among acute stroke patients led to a reduction of NV-HAP (Grossmann et al., 2021; Mitchell et al., 2019). A nurse-led intervention on bedside screening and a rapid clinical swallow undertaken by a speech pathologist led to statistically significant decrease in NV-HAP rates from 6.5% to 2.8%, $p = <0.001$ (Mitchell et al., 2019).

Overall, the growing body of evidence showed the effectiveness of the use of a standardized oral care protocol in reducing the incidence of NV-HAP. Nurses are in the best position as front-line staff to prevent a largely preventable healthcare complication of HAI. To promote holistic patient care, nursing staff education should focus on the importance of adherence to a standardized oral care protocol and delivery in hospitalized patients. The information gathered from the qualitative studies contributes to knowledge acquired about patient-centered care. The perceived values of oral care and acknowledgment by nursing staff would promote high-quality, safe patient care. Due to the study findings and limitations presented in the studies, this doctoral proposal project will use a standardized oral care protocol

to evaluate effectiveness in preventing NV-HAP occurrences in an acute neuro-intensive care setting.

Methods

There is a growing body of evidence that shows the use of an oral care protocol effectively controls the rates of NV-HAP incidences. Oral care is recognized as the most common modifiable risk factor for NV-HAP prevention. Nursing educational support with regular use of a comprehensive oral care protocol are approaches that promote high-quality and safe patient care. This DNP project aims to develop a standardized oral care protocol and will evaluate its implementation to reduce the incidence of NV-HAP.

Design

The quality improvement (QI) project applied the Iowa Framework to plan, develop, implement, and evaluate the effectiveness of the oral care protocol. The QI project used a retrospective comparative outcome three months pre-intervention from September 2022 to November 2022 to establish a baseline. The project was implemented from December 2022 to February 2023. The analysis was conducted to determine the incidence of NV-HAP and the effectiveness of the oral care protocol. The dependent variable in the project was the incidence of NV-HAP. The independent variable in the project was the implementation of a standardized oral care protocol (Appendix B, Figure 2). A measurement instrument used in this project was pre- and post-implementation surveys. A standardized oral care protocol was implemented during the pilot period on all acute cerebral vascular accident (CVA) non-mechanically ventilated patients admitted to the neuro-ICU.

In this DNP project, aggregate patient data regarding the development of NV-HAP was recorded. Each medical diagnosis of NV-HAP was considered one entry of incidence. The pre-intervention baseline data were compared to the post-intervention collected from December 2022

to January 2023. The comparison data will determine if the standardized oral care protocol has had an effect on reducing the incidence of NV-HAP from baseline.

NV-HAP Intervention Audit Tool

The nurse champion recorded the frequency and type of oral care delivered in a 12-hour shift and at bedtime by conducting retrospective chart reviews (Appendix C, Figure 3). Direct patient and family interviews were gathered to determine the frequency of oral care provided. Staff were observed when implementing the oral care protocol to ensure the standard procedures were followed. Patient and nursing barriers to oral care delivery were identified and documented to assist in expanding the protocol to other departments in the hospital. The barriers to compliance were: (a) lack of time, (b) lack of resources to perform routine oral care, (c) not perceived as a priority, (d) lack of oral care supplies, and e) documentation difficulties in the electronic medical record (EMR).

Setting

The facility in which the project took place is a hospital located in Southern California. It is a Magnet-certified center for nursing excellence and is accredited by The Joint Commission. The hospital has 412 beds and is identified as a not-for-profit community medical center and trauma center. The project took place in the 12-bed Neuro ICU. The Neuro ICU is one of four intensive care units. A standardized oral care protocol was developed and tested through a pilot project in the Neuro ICU.

The nurse-driven standardized oral care protocol provided guidance in the level of care required for a patient in three categories: independent (low risk of aspiration), dependent for oral care (moderate risk of aspiration), and completely dependent for oral care (high risk for aspiration). Independent patients will receive a toothbrush and toothpaste, alcohol-free antiseptic

mouth rinse to be used four times daily. Dependent patients that may require assistance with oral care will receive an alcohol-free antiseptic mouth rinse and manufacturer-package oral care kit, including suction toothbrush, anti-plaque oral cleansing solution, and mouth moisturizer.

Completely dependent patients will receive the manufacture-package oral care kit. Oral care for the dependent patients with moderate and high- risk of aspiration will be delivered four times daily.

Participants

A convenience sampling of nurses was conducted in the Neuro-ICU unit. Potential subjects were recruited by e-mail and in person. If the method of recruitment was done by e-mail, an information cover sheet and copy of the pre-implementation surveys were attached and sent to the nurses if they indicated interest. For the in-person contact method with the nurses, the DNP student provided a copy of the pre-implementation survey, a script, and an informational cover letter to describe the intent of the survey. The total number of participants was 40 RNs in the pre-implementation period. More than three-quarters (85%) of RNs had attained a Bachelor's degree or higher and are female (80%). Most participants are Asian (45%), followed by White (27.2%), Hispanic (21.8%), and African American (5.45 %). There were 25% of male gender and 75% of female gender.

Institutional Letter of Support and Institutional Review Board Approval

The project was submitted for approval by the Institutional Review Board (IRB) at California State University, Los Angeles, and the project hospital. A letter of approval was obtained from the project hospital's administrative, clinical executive (see Appendices D, E & F). Patient data was gathered in aggregate, and no personal patient information was obtained for the project.

Planning the Quality Improvement Program

Develop an Implementation Plan

Once the IRB approval was granted, the pilot project began, guided by the Iowa Model. This DNP project was triggered by lack of a standardized oral care protocol in the organization to prevent NV-HAP. The practice change is aligned with the organization's top priority to prevent hospital-acquired infection by providing high-quality, cost-effective, and safe care. The evidence-based literature research was gathered by developing the PICOT question, "does the use of an oral care protocol over two months period decrease the occurrence of NV-HAP by 25% versus usual care among acute stroke non-ventilated patients in a 12-bed Neuro ICU?" The purpose of the project is to evaluate the effectiveness of a standardized nurse driven oral care protocol in the improvement of NV-HAP incidences.

The QI interdisciplinary team that was formed consisted of the APN, nurse champion, critical care physician champion, and infection preventionist. The QI team selected, critiqued, and synthesized the best available evidence. Additionally, the QI team oversaw the pilot project implementation process and evaluated the practice change.

Following the decision to implement the oral care protocol, the QI interdisciplinary team moved forward with the two-month pilot testing of the standardized oral care protocol. The Iowa Model consists of 4 implementation phases (see Appendix G, Figure 4). The 4 phases are adapted from Rogers' Diffusion of Innovation Theory for use in healthcare. Phases 1 and 2 will be conducted before the "go live" begins to prepare clinicians for the implementation of the oral care protocol. In Phase 3, active adoption of the practice change begins with the "go live." Each of the phases will be adopted and discussed separately.

Phase 1 Create Awareness and Interest

Full disclosure of the pilot project was communicated to the nursing management team, clinical, administrative executive, ICU RNs, certified nursing assistants (CNAs), medical staff, speech therapist, physical therapist, infection preventionist, and lastly, to the nurse representatives from the Professional Research Council. The ICU 3 nursing team received information through electronic messages, staff, and huddle meetings conducted on both shifts. Furthermore, announcements and unit-based in-services were provided using educational handouts and posters.

Phase 2- Build Knowledge and Commitment

The nurse champion provided unit-based education opportunities for the RNs, licensed vocational nurse and CNAs. A survey was conducted for the nursing staff to assess their knowledge of the oral care protocol pre-implementation and post-implementation (see Appendix H and I). Education was provided to the nursing team via PowerPoint slides that included the classification of HAP, importance of NV-HAP prevention, risk factors, the Joint Commission call to action recommendations for NV-HAP prevention, and the evidence-based oral care protocol. Nursing management was informed of the new pilot project practice change. The project author and nurse champion met with the Information Technology representative to develop an NV-HAP prevention bundle documentation form in the EMR. The EMR-modified NV-HAP documentation form was included in the nursing education plan.

Phase 3- Promote Action and Adoption

This phase began with the go-live implementation. Reminders of oral care protocol were communicated daily on all shifts. A demonstration of the workflow was outlined and communicated to the nursing staff. The project author collaborated with the nurse champion and

rounded weekly twice a week on both shifts to inform the nurses about the pilot project, address concerns and answer questions.

Patients were included in the oral care protocol if they were 18 years or older and had had an acute CVA, were not mechanically ventilated, and did not have community-acquired pneumonia present on admission. All patients who developed hospital-acquired pneumonia greater than 48 hours were included in the study. Patients less than 18 years of age and admitted without a diagnosis of acute CVA were excluded. Additionally, patients diagnosed with ventilator-acquired pneumonia were excluded. Patients with a length of stay of less than 48 hours and those who had mechanical ventilation removed in less than 48 hours were excluded from the study.

Aggregate patient data were collected on patients who developed a diagnosis of HAP greater than 48 hours after hospital admission. The data were collected at baseline and then during the implementation of the project. If patient is selected, then the RN approached the patient and provided the education flyer to the patient and family member with the information on the oral care protocol (see Appendices J, K, Figure 5 & 6). The trained day shift RN on the oral care protocol performed observations during the day shift, at mealtimes, twice daily. Oral care frequency, assessment, and type of oral care method were recorded on the NV-HAP intervention audit tools on both shifts by trained auditors. Random observations of the nursing staff were conducted to evaluate fidelity to the oral care protocol. Any barriers were recorded on the NV-HAP intervention audit tool. The International Classification of Diseases 10 codes for NV-HAP diagnosis were used to identify the diagnosis of NV-HAP (Appendix L) . Resource educational materials and quick reference guides of the oral care protocol were available for the

nursing staff. Trained RNs and project study author reported the progress and updates of the pilot project at unit-based huddles and staff meetings.

Phase 4- Pursue Integration and Sustained Use of the Oral Care Protocol

This phase ends the post-test implementation of the pilot. The project author will provide a report of NV-HAP incidences and the results of the survey responses to the key stakeholders. Report of audits, feedback including barriers, will be provided to the nursing team and nursing management. The oral care protocol will be evaluated and revised from feedback from the nursing staff. QI project findings will be reported to the Clinical Practice Council, Professional Research Council, and Ventilator-Associated Pneumonia Committees.

Measures

The first measure used in this quality improvement research study was pre and post-implementation surveys to assess nurses' knowledge of an oral care protocol to prevent non-ventilator hospital-acquired pneumonia. The survey used is a validated tool adapted with permission from Diane Baker with a resource toolkit provided to conduct this study. The original survey tool was used by Feider and Mitchell (2010) to evaluate nurses' knowledge of an oral care protocol for mechanically ventilated patients. Outcome measures of NV-HAP incidents were used to determine whether there was an improvement from the education that was provided to nurses regarding the oral care protocol.

Data Collection and Management

Two surveys were provided per nurse who voluntarily participated in the study. The survey completion was not mandatory. The nurse champion and nurse manager provided the participants with the pre-implementation surveys that were completed prior to the beginning of the pilot study implementation. At the completion of the study, a post-implementation survey

was provided in person by the RN champion and nurse manager. Data were collected for two weeks at the end of the pilot study. Responses to the survey were kept in a locked computer located in a locked office when the computer was not in use. The project author only had access to the computer where the survey responses were stored.

Baseline data collection on the incidence of NV-HAP was collected over a 3-month period from September -November 2022. Admission data from December 2022 - February 2023 was collected for a pre-intervention and post-intervention comparison of NV-HAP incidences.

Decision support provided aggregate patient data of the NV-HAP discharge diagnosis using the International Classification of Diseases (ICD) 10 codes. De-identified patient data were collected and entered in an Excel spreadsheet and stored in a password-protected locked computer that was accessed only by the project author. The computer was located at the project setting in a locked office.

Results

Data Analysis

There was a total of 45 staff RNs employed in the Neuro ICU unit before the pilot study began in December 2022. Forty (91%) survey responses were collected prior to the implementation of the oral care protocol, and 33 (75%) survey responses were collected within two weeks at the end of the pilot study. The survey data was transferred to an Excel spreadsheet for analysis. Data analysis was performed using the Excel application and statistical features. The Chi-square test was used to compare the nurses' knowledge assessment of the NV-HAP diagnosis in the pre and post-implementation of the protocol and understanding of the importance of the oral care protocol before the start of the pilot study and after the implementation of the oral care protocol.

Results showed an 18.8% increase in nurses' knowledge of the NV-HAP diagnosis post-implementation (78.8%) from pre-implementation (60.0%) of an oral care protocol. The results showed no statistical significance ($p = 0.085$). Although there was an increase in nurses' understanding of the importance of using an oral care protocol to prevent NV-HAP pre-implementation (42.5%) and post-implementation (46.9%), no statistical significance difference ($p = 0.304$) was found (Appendix M).

The nursing frequency of oral care was performed using the Chi-square. It showed oral care was mostly done every four hours in the pre-implementation (72.5%) and post-implementation (87.5%). The results showed no statistical significance between the baseline data and post-implementation data ($p = 0.824$).

Nurses consistently used the oral care kit in both the pre-implementation (72.5%) and post-implementation (87.5%) but showed no statistical significance ($p = 0.119$).

Although results showed an increase in nursing performance of an oral care assessment for their patients from pre-implementation (70%) to post-implementation (91%) periods, Chi-square test results did not show statistical significance, $\chi^2 (2, N= 73) = 5.21, p= 0.073$.

Nurses performed an oral care assessment and documented it in the electronic medical record every four hours pre-implementation (42.5%) and once a shift post-implementation (48.5%). The Chi-square performed showed that there was no statistically significant difference between the two time periods, $\chi^2 (4, N=73) = 2.87, p = 0.579$.

Lastly, this project investigated RN's perception of an oral care protocol in the prioritization of nursing care. Data showed nurses placed oral care protocol as a low priority in the pre-implementation period (45%) and rated it a moderate priority in the post-implementation period (48.5%), and no statistical significance among the two periods $\chi^2 (2, N73) = 3.51, p = 0.172$), despite the intervention (Appendix O). The nursing team may have inadvertently placed oral care as low priority due to the competing demands and needs of critically ill patients. The certified nursing assistants who mostly assisted the nurses in providing oral care did not routinely communicate with the RNS after oral care was provided. This led to identification of missing oral care and inconsistent documentation in the EMR. Post-implementation of the oral care protocol, the nurses and nursing assistants not only performed oral four times a day but increased the duration of toothbrushing up to two minutes (Appendix O). There was no report of NV-HAP incidences during the two months pilot project compared to two NV-HAP incidences at baseline.

Discussion

Nurses' knowledge of the NV-HAP diagnosis and understanding increased post-implementation regarding the importance of an oral care protocol in the prevention of NV-HAP. Pre-implementation, there were two NV-HAP incidences, and post-implementation, there were none, suggesting that implementing a nurse-driven oral protocol can decrease the incidence of NV-HAP. These findings are consistent with the quality improvement initiatives performed by other researchers (Schutte & Warren, 2020). Schutte and Warren (2020) identified improved nurse knowledge following the implementation of an oral care protocol.

In Table 2, the staff nurses' knowledge improved regarding the importance of providing patients with an oral care protocol; however, this increase was not significant. This may have been due to attrition of the participant; there were 40 participants pre-implementation versus 33 post-implementation. Attrition was due to several factors, such as medical leaves, extended vacations, or leaving the organization. Resource nurses were utilized to fill in the gaps when there was insufficient staff on the unit. The resource nurse,s as a group, had not all received the education. Additionally, nurses in all ICU departments were familiar with the preventive strategies for VAP, a subset of HAP.

The nursing staff in the Neuro ICU had strong foundational knowledge of the importance of routine oral care practices for HAP prevention. Interestingly, it is reflected in the nurse's pre and post-survey responses that there was consistency in the frequency of oral performed every four hours in 24 hours. Moreover, there exists a collaborative practice among the nurse and respiratory therapist in the provision of evidence-based VAP bundled practices that includes the oral care protocol. Outcomes data and adherence to this protocol are monitored and reported regularly by the VAP task force committee. The pre-existing knowledge of oral care practices

relative to infection prevention and a lower sample of surveys post-implementation may have likely contributed to the lack of statistical significance in the current project's results.

Interestingly after the nurse champion and study author provided an initial 30-minute lecture and additional unit-based in-services, there was a clinically significant increase in knowledge of the NV-HAP diagnosis. This increase in awareness may potentially enhance the nurses' understanding of the importance of providing oral care to high-risk non-vented acute stroke patients.

Overall, nurses' perception of the prioritization of oral care compared to other nursing tasks improved to moderately important compared to low priority at baseline. A paradigm shift in the nurses' improvement in the prioritization of oral care was due to staff re-education and engaging patients and families in education on the new oral care protocol. The collaborative practices between the RN and CNA were strengthened as the perception of oral care as optional nursing care for patient's comfort but an important evidence-based infection control practice.

Limitations

Several limitations and opportunities were identified. The study was conducted in a single small acute hospital intensive care unit of 12 beds. Generalizability was limited to other hospitals of similar hospital bed sizes. There were a higher number of travel nurses who filled in the resource needs in the ICU during the winter months leading to the variability and probable inconsistencies of oral care practices and nursing documentation in the EMR. The RN champion only worked during the day shift, which hindered chart audits during the night shift. The duration of pilot project took place for over two months, limiting the observation of staff fidelity to the oral care protocol and outcome measures of NVHAP rates. Some patients refused the oral care protocol despite patient and family education. The project author was unable to link the pre- and

post-surveys to the individual responses leading to an independent group analysis of the survey responses. Additionally, demographic data of the nursing staff was lacking in the post-implementation survey.

Recommendations

A nurse-driven evidence-based practice of oral care protocol is fundamental nursing care that is often missed. In this project, leadership support, education, and adequate oral care supplies within reach were successful in improving adherence to an oral care protocol. Identification of potential barriers prior to the start of the oral care intervention is suggested to avoid non-adherence to the oral care protocol and mitigate risks for NV-HAPs. Support by nursing leadership, open communication, and discussion of issues are necessary to identify gaps and potential solutions in the processes. Nurses, as front-line staff, are in the best position to communicate effectively with patients and family members regarding the importance of oral care. Strengthening the nurse-patient relationship is a patient-centered approach, which is key to the active participation of patients and family members in performing self-care activities. Ultimately improvement in communication and self-care activities contributed not only to performance improvement metrics but also positively impacted customer satisfaction scores. Including the certified nursing assistants as valuable members of the nursing team are empowering and may lead to adherence to the oral care protocols. Additionally, given the multiple tasks required of nurses, delegation of oral care to CNA could ensure that it is consistently provided. Lastly, creating an organizational dashboard to track and trend performance of NV-HAP rates and clinical outcomes is important to evaluate the effectiveness of the oral care intervention and maintain sustainability.

Conclusion

The results of this quality improvement project showed that nurses demonstrated improved knowledge of the importance of an oral care protocol in preventing NV-HAP after receiving education and training. The results of this quality improvement project aligned with the findings found in the literature (Guiliano et al., 2021; Jenson et al., 2018; Schutte & Warren, 2020, Warren et al., 2019). Additionally, there was a clinical reduction of NV-HAP incidences; thus, the findings suggest that a standardized oral care protocol may be effective in NV-HAP prevention. Key elements to the successful implementation and adherence of the oral care protocol were accomplished through having trained nursing staff with enhanced education, providing clinical resources and oral care supplies, conducting weekly tracking of compliance through EHR chart audits, and nursing leadership support.

Interestingly, mandatory nursing education alone was not enough to impact compliance. Partnership with patients and family members was equally important in implementing the NV-HAP prevention strategy using oral care. Moreover, patients and family members appreciated nurses offering help with oral care and ensuring oral care hygiene was adequately done; this is similar to the findings in the literature (Coker et al., 2020). Active tracking of compliance and communication of results with the nursing management team promoted staff accountability. Nursing education of the certified nursing aides and licensed vocational nurses was empowering, ensuring they were active participants in the delivery of standardized oral care to the patients so their needs are met. Although the results of this project are motivating, it should be noted that there are limitations of the project, which were short duration conducted on the pilot project of two months, and a small sample size of study participants.

The results showed that the modifiable risk factor of pneumonia through a nurse-driven standardized oral care protocol might be an effective strategy in NV-HAP prevention. Future research should focus on the duration and frequency of oral care frequency and standardized implementation of oral care programs in other hospital non-critical care units through a large RCT. Although NV-HAPs are avoidable, achieving zero HAI may be challenging to reach. Prevention of NV-HAPs should be an organizational priority, which can be accomplished by increasing awareness and implementing preventive strategies in an acute hospital setting.

References

- Baker, D., & Quinn, B. (2018). Hospital acquired pneumonia prevention initiative-2: Incidence of nonventilator hospital-acquired pneumonia in the United States. *American Journal of Infection Control*, 46(1), 2-7. <https://doi.org/10.1016/j.ajic.2017.08.036>
- Boden, I., Skinner, E. H., Browning, L., Reeve, J., Anderson, L., Hill, C., Robertson, I. K., Story, D., & Denehy, L. (2018). Preoperative physiotherapy for the prevention of respiratory complications after upper abdominal surgery: Pragmatic, double blinded, multicenter randomized controlled trial. *BMJ*, j5916. <https://doi.org/10.1136/bmj.j5916>
- Brown, C. G. (2014). The Iowa Model of Evidence-Based Practice to promote quality care: an illustrated example in oncology nursing. *Clinical Journal of Oncology Nursing*, 18(2), 157-159. <https://doi.org/10.1188/14.cjon.157-159>
- Buckwalter, K. C., Cullen, L., Hanrahan, K., Kleiber, C., McCarthy, A. M., Rakel, B., Steelman, V., Tripp-Reimer, T., & Tucker, S. (2017). Iowa model of evidence-based practice: Revisions and validation. *Worldviews on Evidence-Based Nursing*, 14(3), 175-182. <https://doi.org/10.1111/wvn.12223>
- Chick, A., & Wynne, A. (2020). Introducing an oral care assessment tool with advanced cleaning products into a high-risk clinical setting. *British Journal of Nursing*, 29(5), 290-296. <https://doi.org/10.12968/bjon.2020.29.5.290>
- Coker, E., Ploeg, J., & Kaasalainen, S. (2020). Relying on nursing staff for oral hygiene care: A qualitative interpretive description study. *Geriatric Nursing*, 41(6), 891-898. <https://doi.org/10.1016/j.gerinurse.2020.06.015>
- Centers for Disease Control and Prevention. (2016). Prevention status report, healthcare associate infections.

<https://www.cdc.gov/hai/pdfs/progress-report/2016-Progress-Report-Executive-Summary-H.pdf>

Cullen, L., Hanrahan, K., Farrington, M., DeBerg, J., Tucker, S., & Kleiber, C. (2018). *Evidence-based practice in action: Comprehensive strategies, tools, and tips from the University of Iowa hospitals and clinics*. Sigma Theta Tau International.

Davis J., & Finley, E. (2012). The breadth of hospital-acquired pneumonia; non-ventilated versus ventilated patients in Pennsylvania. *Patient Safety Advisory*, 2012(9), 99-105.

https://digirepo.nlm.nih.gov/master/borndig/101591716/201209_99.pdf

Ferguson, C., George, A., Villarosa, A. R., Kong, A. C., Bhole, S., & Ajwani, S. (2020).

Exploring nursing and allied health perspectives of quality oral care after stroke: A qualitative study. *European Journal of Cardiovascular Nursing*, 19(6), 505-512.

<https://doi.org/10.1177/1474515119886798>

Feider, L. L., Mitchell, P., & Bridges, E. (2010). Oral care practices for orally intubated critically ill adults. *American Journal of Critical Care*, 19(2), 175-183.

<https://doi.org/10.4037/ajcc2010816>

Fine, L. S. (2020). Non-ventilator health care-associated pneumonia (NV-HAP): Pathogenesis and microbiology of NV-HAP. *American Journal of Infection Control*, 48(5), A7-A9.

<https://doi.org/10.1016/j.ajic.2020.03.009>

Giuliano, K. K., Baker, D., & Quinn, B. (2018). The epidemiology of nonventilator hospital-acquired pneumonia in the United States. *American Journal of Infection Control*, 46(3),

322-327. <https://doi.org/10.1016/j.ajic.2017.09.005>

- Giuliano, K. K., Penoyer, D., Middleton, A., & Baker, D. (2021). Original research: Oral care as prevention for non-ventilator hospital-acquired pneumonia. *AJN, American Journal of Nursing*, 121(6), 24-33. <https://doi.org/10.1097/01.naj.0000753468.99321.93>
- Grossmann, I., Rodriguez, K., Soni, M., Joshi, P. K., Patel, S. C., Shreya, D., Zamora, D. I., Patel, G. S., & Sange, I. (2021). Stroke and pneumonia: Mechanisms, risk factors, management, and prevention. *Cureus*.
- Grove, S. K., Gray, J. R. 2019). Building an evidence-based nursing practice. In S. K. Grove (ed.). *Understanding nursing research: Building an evidence-based practice* (7th ed.) (pp. 387-429). Elsevier.
- Jenson, H., Maddux, S., & Waldo, M. (2018). Improving oral care in hospitalized non-ventilated patients: Standardizing products and protocol. *Medsurg Nursing*, 27(1), 38-45.
<https://www.proquest.com/openview/9fd75da460233e3cc953252fd67ecaa3/1?pq-origsite=gscholar&cbl=30764>
- Khasanah, I. H., Sae-Sia, W., & Damkliang s, J. (2019). The effectiveness of oral care guideline implementation on oral health status in critically ill patients. *SAGE Open Nursing*, 5, 1-9
<https://doi.org/10.1177/2377960819850975>
- Klompas, M. (2022). Risk factors and prevention of hospital-acquired and ventilator associated pneumonia in adults. *UptoDate*.
<https://www.uptodate.com/contents/treatment-of-hospital-acquired-and-ventilator-associated-pneumonia-in-adults>
- Lacerna, C. C., Patey, D., Block, L., Naik, S., Kevorkova, Y., Galin, J., Parker, M., Betts, R., Parodi, S., & Witt, D. (2020). A successful program preventing nonventilator hospital-

- acquired pneumonia in a large hospital system. *Infection Control & Hospital Epidemiology*, 41(5), 547-552. <https://doi.org/10.1017/ice.2019.368>
- Lukasewicz Ferreira, S., Hubner Dalmora, C., Anziliero, F., De Souza Kuchenbecker, R., & Klarmann Ziegelmann, P. (2022). Factors predicting non-ventilated hospital-acquired pneumonia: Systematic review and meta-analysis. *Journal of Hospital Infection*, 119, 64-76. <https://doi.org/10.1016/j.jhin.2021.09.024>
- Micek, S. T., Chew, B., Hampton, N., & Kollef, M. H. (2016). A case-control study assessing the impact of Nonventilated hospital-acquired pneumonia on patient outcomes. *Chest*, 150(5), 1008-1014. <https://doi.org/10.1016/j.chest.2016.04.009>
- Magill, SS., O’Leary E. Janelle S. J., Thompson, D.L., Dumyati, G., Nadle, J., Wilson, L. E., Kainer M.A., Lynfield, R., Greissman, S., Ray, S, M., Beldavs Z., Gross, C., Bambwer, W., Sievrs, M., Concannon, N., Buhr, L., Warnke, M., Maloney, M.,... Edwards, J. R. (2018). Changes in prevalence of health care–associated infections in US Hospitals. (*New England Journal of Medicine*, 379(11), 1732-1744. <https://doi.org/10.1056/NEJMoa1801550>
- Mitchell, B. G., Russo, P. L., Cheng, A. C., Stewardson, A. J., Rosebrock, H., Curtis, S. J., Robinson, S., & Kiernan, M. (2019). Strategies to reduce non-ventilator-associated hospital-acquired pneumonia: A systematic review. *Infection, disease & health*, 24(4), 229–239. <https://doi.org/10.1016/j.idh.2019.06.002>
- Munro, S., & Baker, D. (2018). Reducing missed oral care opportunities to prevent non-ventilator-associated hospital-acquired pneumonia at the Department of Veterans Affairs. *Applied Nursing Research*, 44, 48-53. <https://doi.org/10.1016/j.apnr.2018.09.004>

- Munro C. S., Baker, D., Giuliano K. K., Sullilvan S. C., Haber, J., Jones, B.E., Crist, M. B., Nelson, R. E., Carey, E., Lounsbury, O., Lucatorto, M., Miller, R., Pauley. B., & Klompas, M. (2021). Nonventilator hospital-acquired pneumonia: A call to action. *Infection Control & Hospital Epidemiology*, 2021(42), 991-996.
<https://doi.org/10.1017/ice.2021.239>
- Nabeya, D., Kinjo, T., Parrott, G. L., Uehara, A., Motooka, D., Nakamura, S., Nahar, S., Nakachi, S., Nakamatsu, M., Maeshiro, S., Haranaga, S., Tateyama, M., Tomoyose, T., Masuzaki, H., Horii, T., & Fujita, J. (2017). The clinical and phylogenetic investigation for a nosocomial outbreak of respiratory syncytial virus infection in an adult hemato-oncology unit. *Journal of Medical Virology*, 89(8), 1364-1372.
<https://doi.org/10.1002/jmv.24800>
- Pritts, K., Kerber, C., Calderon, S., & Bigler, P. (2020). Implementation of a Standardized Oral Care Protocol to Decrease Non-Ventilator Hospital-Acquired Pneumonia. *Medsurg Nursing*, 29(6), 387-390.
- Quinn, B., Baker, D. L., Cohen, S., Stewart, J. L., Lima, C. A., & Parise, C. (2014). Basic nursing care to prevent Nonventilator hospital-acquired pneumonia. *Journal of Nursing Scholarship*, 46(1), 11-19. <https://doi.org/10.1111/jnu.12050>
- Saied, W. I., Mourvillier, B., Cohen, Y., Ruckly, S., Reignier, J., Marcotte, G., Siami, S., Bouadma, L., Darmon, M., De Montmollin, E., Argaud, L., Kallel, H., Garrouste-Orgeas, M., Soufir, L., Schwebel, C., Souweine, B., Glodgran-Toledano, D., Papazian, L., ... Timsit, J. (2019). A comparison of the mortality risk associated with ventilator-acquired bacterial pneumonia and Nonventilator ICU-acquired bacterial pneumonia*. *Critical Care Medicine*, 47(3), 345-352. <https://doi.org/10.1097/ccm.0000000000003553>

- Satheeshkumar, P. S., Papatheodorou, S., & Sonis, S. (2020). Enhanced oral hygiene interventions as a risk mitigation strategy for the prevention of non-ventilator-associated pneumonia: A systematic review and meta-analysis. *British Dental Journal*, 228(8), 615-622. <https://doi.org/10.1038/s41415-020-1452-7>
- Schutte, D. L., & Warren, C. (2020). Staff Knowledge and Attitudes toward Oral Care in an Acute Care Hospital Pre-and Post-Implementation of an Evidence-Based Practice Change. *Medsurg Nursing*, 29(3)
- Tang, X., Shen, Y., Pan, X., Liao, J., Xu, Y., Luo, W., Zhang, X., Li, C., Wan, Q., Cai, X., Zhang, X., Wang, T., & Zhang, G. (2020). Oral care for intensive care unit patients without mechanical ventilation: Protocol for a systematic review and meta-analysis. <https://doi.org/10.21203/rs.3.rs-121895/v1>
- Talley, L., Lamb, J., Harl, J., Lorenz, H., & Green, L. (2016). HAP prevention for nonventilated adults in acute care. *Nursing Management*, 47(12), 42-48. <https://doi.org/10.1097/01.numa.0000508259.34475.4c>
- Tesoro, M., Peyser, D. J., & Villarente, F. (2018). A retrospective study of non-ventilator-associated hospital acquired pneumonia incidence and missed opportunities for nursing care. *Journal of Nursing Administration*, 48(5), 285-291. <https://doi.org/10.1097/01.nna.00000000000000614>
- Titler, M. G., Kleiber, C., Steelman, V. J., Rakel, B. A., Budreau, G., Everett, L. Q., Buckwalter, K. C., Tripp-Reimer, T., & Goode, C. J. (2001). The Iowa model of evidence-based practice to promote quality care. *Critical Care Nursing Clinics of North America*, 13(4), 497-509. [https://doi.org/10.1016/s0899-5885\(18\)30017-0](https://doi.org/10.1016/s0899-5885(18)30017-0)

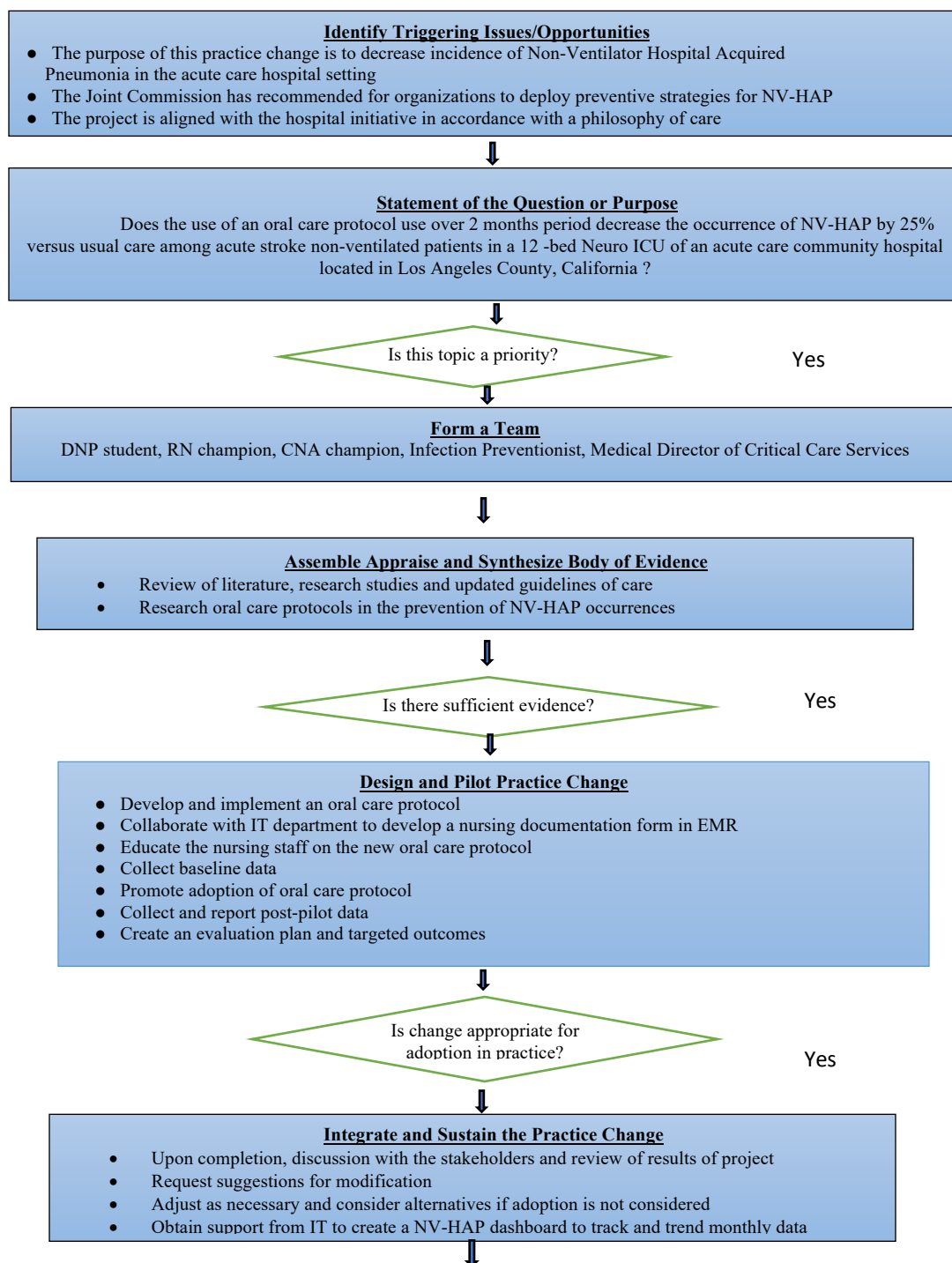
- Tucker, S., McNett, M., Mazurek Melnyk, B., Hanrahan, K., Hunter, S. C., Kim, B., Cullen, L., & Kitson, A. (2021). Implementation science: Application of evidence-based practice models to improve healthcare quality. *Worldviews on Evidence-Based Nursing*, 18(2), 76-84. <https://doi.org/10.1111/wvn.12495>
- The Joint Commission. (2021). Quick Safety. *Preventing non-ventilator hospital acquired pneumonia*. <https://www.jointcommission.org/resources/news-and-multimedia/news/2021/09/new-quick-safety-on-preventing-nvhap/>
- Vanhems, P., Bénét, T., & Munier-Marion, E. (2016). Nosocomial influenza. *Current Opinion In Infectious Diseases*, 29(4), 366-372. <https://doi.org/10.1097/qco.0000000000000287>
- Warren, C., Medei, M. K., Wood, B., & Schutte, D. (2019). A nurse-driven oral care protocol to reduce hospital-acquired pneumonia. *AJN, American Journal of Nursing*, 119(2), 44-51. <https://doi.org/10.1097/01.naj.0000553204.21342.01>
- Yuan, D., Zhang, J., Wang, X., Chen, S., & Wang, Y. (2020). Intensified oral hygiene care in stroke-associated pneumonia: A pilot single-blind randomized controlled trial. *INQUIRY: Journal of Health Care Organization, Provision, and Financing*, 57, 1-7 <https://doi.org/10.1177/0046958020968777>
- Zhao, J., Duan, S., Liu, X. Han, L. Jiang, Y., Wang, J., Gao, S, & Hao, Y., Y. (2016). Discussion on clinical application of Iowa model in TCM nursing care. *Alternative, Complementary & Integrative Medicine*, 2(1), 1-5. <https://doi.org/10.24966/acim-7562/100011>

Appendix A

Iowa Model

Figure 1

Iowa Model-Revised: Evidence-Based Practice to Promote Excellence in Health Care



Disseminate Results

Contact publications to request acceptance of manuscript

Note. Used/reprinted with permission from the University of Iowa and Clinics, copyright 2015.

Appendix B

Oral Care Protocol

Figure 2

Standardized Oral Care Protocol

Self-care and staff-assist. Able to expectorate (spit)  Low risk for aspiration	EQUIPMENT • Soft toothbrush, ADA approved • Toothpaste, ADA approved • Alcohol free antiseptic mouthwash • Mouth moisturizer prn • Lip balm (optional) FREQUENCY • After each meal and before bedtime. • If patient is NPO, oral care should be done 4 times daily.	PROCEDURE 1. Set patient up at sink or in bed with all equipment. 2. Instruct patient to brush teeth for 1-2 minutes. 3. Instruct to use antiseptic oral rinse to swish for at least 30 seconds. 4. May moisturize interior of mouth and lips using an oral swab prn 5. Discard disposable equipment/swab in appropriate receptacle.
Dependent for oral care. Not able to expectorate (spit or swallow). Moderate risk of aspiration 	EQUIPMENT • Suction toothbrush with anti-plaque oral cleaning solution packet (available in the kit) • Suction tubing and cannister • Alcohol-free antiseptic oral rinse • Mouth moisturizer • Lip balm (optional) FREQUENCY • After each meal and before bedtime. • If patient is NPO, oral care should be done 4 times daily.	PROCEDURE 1. Reposition HOB to at least >30 degrees unless contraindicated. 2. Prepare equipment at bedside. 3. Moisten suction toothbrush or regular toothbrush in anti-plaque oral rinse. 4. Connect suction toothbrush to suction device 5. Assist the patient to brush all surfaces of the teeth until clean for 1-2 minutes. 6. Suction debris from mouth. 7. Apply mouth moisturizer using oral swab. To the interior of the oral cavity and apply lip balm. 8. Discard disposable equipment in appropriate receptacle.
Completely dependent on oral care, failed dysphagia screen and swallow evaluation High Risk of aspiration 	EQUIPMENT • Suction toothbrush/ or suction swab and oral cleansing solution (available in the kit) • Oral cleansing solution in toothbrush kit • Mouth moisturizer FREQUENCY • After each meal and before bedtime.	PROCEDURE 1. Reposition HOB to at least >30 degrees unless contraindicated. 2. Prepare equipment at bedside. 3. Perform suction as needed to remove oropharyngeal secretions 4. Obtain suction toothbrush/oral swab and moisten with oral anti-plaque cleansing solution. 5. Moisten suction toothbrush with oral cleansing solution. 6. Connect suction toothbrush to suction device. 7. Suction debris from mouth clean gums and tongue 8. Apply moisturizer using oral swab to the interior of the oral cavity and lips.
Denture care of patients with no teeth. Before the patient goes to sleep, remove and clean dentures and place them in a denture cleansing solution once daily. 	EQUIPMENT • Denture cup, labeled • Denture brush is preferred when available, otherwise soft toothbrush • ADA approved denture cleanser (for soaking) • 2 oral swabs • Denture adhesive (optional) • Mouth moisturizer prn or mouthwash FREQUENCY • Dentures are removed for cleaning at bedtime.	PROCEDURE 1. After removing dentures, place in a labeled denture cup. 2. Brush the palate, buccal surfaces, gums, and tongue with the toothbrush or swab. 3. Patient can swish and spit mouthwash, or use oral swab to apply moisturizer. 4. Clean and dry equipment and return to patient's bedside table. 5. Assist patient in inserting dentures in to mouth. 6. Soak dentures in a denture cleanser in the denture cup. 7. If patient needs denture adhesive to hold firmly in place, follow manufacture directions.

Note. Adapted from “Oral Care as Prevention of Nonventilator Hospital –Acquired Pneumonia: A Four-Unit Cluster Randomized Study,” by K. K. Guiliano, D. Penoyer, A. Middleton, and D. Baker, 2021, *AJN, American Journal of Nursing*, 121(6), <https://doi.org/10.1097/01.naj.0000753468.99321.93>. Adapted with permission.

Appendix C

Oral Care Protocol Audit Tool

Figure 3

Oral Care Protocol Audit Tool

ORAL CARE PROTOCOL AUDIT TOOL										
PATIENTS STICKER										
Date of Audit										
HOB at 30-45 deg.	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a	Y / N n/a
HOB at 30-45 deg. (if n/a place comment)										
Independent/Dependent	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Swallow evaluation	Passed / failed	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Oral Care Charted by RN (QID after meal) 09:00 AM 1:00 PM 8:00 PM 11:00 PM	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Oral care done independently by the patient or family member (QID after meals) 09:00 AM 1:00 PM 8:00 PM 11:00 PM	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Oral Care Suction kit found in room	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Denture Care	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N	Y / N
Comment										

Appendix D

Institutional Review Board Letter for California State University, Los Angeles

Office Memorandum



DATE: November 14, 2022

TO: Raymund Gantioque, DNP

FROM: LaTreshia Scott

PROJECT TITLE: [1961082-1] Non-ventilator Hospital Acquired
Infection Prevention Through Oral Care

REFERENCE #: 22-25X

SUBMISSION TYPE: New Project

ACTION: DETERMINATION OF EXEMPT STATUS

DECISION DATE: November 14, 2022

REVIEW CATEGORY: Exemption category # 3i(A), 3i(B)

Thank you for your submission of New Project materials for this project. The California State University, Los Angeles (Cal State LA) IRB has determined that this project is EXEMPT FROM IRB REVIEW according to federal regulations.

We will retain a copy of this correspondence within our records.

IF ANY CHANGES ARE MADE TO THE METHODS AND PROCEDURES DESCRIBED IN THIS PROTOCOL, YOU MUST SUBMIT A MODIFICATION APPLICATION SO THAT THE PROJECT MAY BE RE-EVALUATED FOR EXEMPTION FROM IRB REVIEW.

For any inquiries about this project, please use the Send Project Mail option on IRBNet to contact LaTreshia Scott. **It allows us to keep all correspondence related to this project in one place, rather than scattered between IRBNet and emails.**

This letter has been electronically signed in accordance with all applicable regulations, and a copy is retained within California State University, Los Angeles (Cal State LA) IRB's records.

Appendix E

Institutional Review Board Letter for Project Hospital



July 27, 2022

Nora Catipon MSN, RN

RE: Our Study # 2022-0711 **At:** Pomona Valley Hospital Medical Center

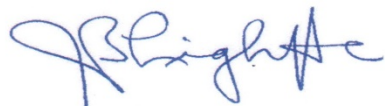
Dear Catipon:

Meeting Date: 7/27/2022
Protocol Title:
Non-ventilator Hospital -Acquired Pneumonia (NVHAP) Prevention Through Oral Care / Study
Number: 2022-0711

This is to advise you that the above referenced Study 2022-0711 has been presented to the Institutional Review Board, and the following action taken subject to the conditions and explanation provided below.

Internal #: New Appl
Expiration Date: 7/26/2023
On Agenda For: Expedited
Description: Date Received- 7/11/2022; Reason Expedited- 21CFR 56.110 (b) (1) - Minimal Risk
Risk Remarks: The problem this project addresses is poor oral health which is associated with a high prevalence of NV-HAP rates in an acute hospital setting in Southern California. NVHAP is one of the leading causes of hospital acquired infections.
IRB ACTION: Approved

Sincerely yours,



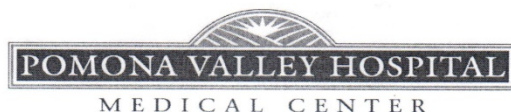
Johnson Lightfoote, M.D.
Chairperson
PVHMC Institutional Review Board

pvhmc.org

1798 North Garey Avenue | Pomona, CA 91767 | 909.865.9500

Appendix F

Letter of Approval to Conduct the Project



Expert care with a personal touch

Lolla Mitchell MSN, RN, NEA-BC

Director of Clinical Practice & Operations

Nursing Administration

4/26/2022

This letter serves as a confirmation of organizational support for Nora B Catipon to perform her DNP project titled Standardized Oral Care Nurse Driven Protocol to Reduce the Incidence of Non-Ventilator Hospital Acquired Pneumonia. This Quality Improvement project will be pilot tested in a 12-bed Neuro-ICU for 3 months period. I commit supporting implementation of this project within Pomona Valley Hospital Medical Center (PVHMC).

Upon obtaining all necessary clearances from the Nursing Research Council at Pomona Valley Hospital Medical Center, and after obtaining the necessary IRB approval, the project lead is allowed to:

1. Collect data from the Electronic Medical Record (EMR). This will included collection of de-identified patient information that have been diagnosed with NV-HAP from admission and up until transfer out of the Neuro-ICU.
2. Conduct necessary interactions with the nursing staff relevant to the project.

This project lead is responsible for ensuring that all activities related to conducting the project are in compliance with the policies that govern practice, HIPPA, research and research –related regulations at PVHMC.

If you have any questions or concerns, please do not hesitate to contact me.

Sincerely,

A handwritten signature in blue ink, appearing to read "Lolla Mitchell", written over a horizontal line.

Lolla Mitchell MSN, RN, NEA-BC

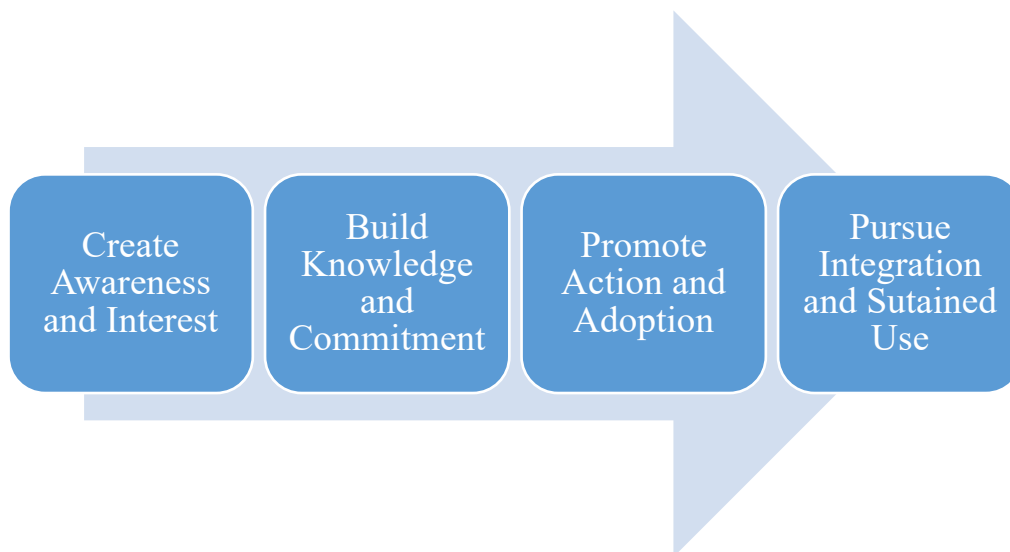
Director of Clinical Practice and Operations

Appendix G

Implementation Phases of the IOWA Model

Figure 4

Implementation Phases



Note. Used/reprinted with permission from the University of Iowa and Clinics, copyright 2015.

Appendix H

Pre-Implementation Survey

Please completely answer each question

1. What is our current position that best describes your position in the Neuro-ICU?

- ☐ Certified Nursing Assistant
- ☐ Licensed Vocational Nurse
- ☐ Registered Nurse

2. How many years have you worked in direct patient care?

- ☐ 0 - 5 years
- ☐ >5- 10 years
- ☐ >10 years- 20 years
- ☐ >20 years

3. How often in a shift do you typically provide oral care? Choose one

- ☐ Every 2 hours
- ☐ Every 4 hours
- ☐ Four times a day after meals
- ☐ Only at bedtime
- ☐ Once daily

4. What type of oral care method do you use when providing care for the neuro patients not on the mechanical ventilator

- ☐ Toothbrush and toothpaste only
- ☐ Oral Care kit available (contains the toothbrush with suction, anti-plaque solution and moisturizer)

5. How much do you know about the medical diagnosis of Non-ventilator Hospital Acquired Pneumonia?

- ☐ Not knowledgeable at all
- ☐ Slightly knowledgeable
- ☐ Moderately knowledgeable
- ☐ Very knowledgeable
- ☐ Extremely knowledgeable

6. How much do you understand about the importance of an oral care protocol in the prevention of hospital acquired pneumonia?

- ☐ Not knowledgeable at all
- ☐ Slightly knowledgeable
- ☐ Moderately knowledgeable
- ☐ Very knowledgeable
- ☐ Extremely knowledgeable

7. What is the typical duration when you provide oral care using a toothbrush or with the toothbrush kit?

- ☐ 1 minute
- ☐ 2 minutes
- ☐ do not pay attention with the time
- ☐ _____

8. Do you perform an oral care assessment prior to providing oral care and document in the electronic medical record?

- ☐ Yes
- ☐ No
- ☐ _____

9. How often do you assess the oral cavity and document in the electronic medical record?

- ☐ Every 4 hours
- ☐ Four times a day
- ☐ Twice a shift
- ☐ Once every shift

10. Given all the nursing task to complete for the care of the acute stroke patient, please rate the priority level of care. Choose one.

- ☐ High priority
- ☐ Moderate priority
- ☐ Low priority

11. What items of the oral cavity do you assess? Check all that apply

- ☐ Redness
- ☐ Swelling
- ☐ Dry Mouth

- ☐ Oral mucosa tear, ulcerations, abrasions, and cracks
- ☐ Lack of saliva
- ☐ Dental carries
- ☐ Bleeding
- ☐ Tenderness
- ☐ Other_____

12. How often do you perform an oral assessment?

- ☐ Once a shift
- ☐ Twice a shift
- ☐ I do not do an oral assessment
- ☐ Other_____

Appendix I

Post Implementation Survey

Please completely answer each question

1. How often in a shift do you typically provide oral care? Choose one

- ☐ Every 2 hours
- ☐ Every 4 hours
- ☐ Four times a day after meals
- ☐ Only at bedtime
- ☐ Once daily
- ☐ _____

2. What type of oral care method do you use when providing care for the Neuro ICU patients not on the mechanical ventilator?

- ☐ Toothbrush and toothpaste only
- ☐ Oral Care kit available (contains the toothbrush with suction, anti-plaque solution and moisturizer)

3. How much do you know about the medical diagnosis of Non-ventilator Hospital Acquired Pneumonia?

- ☐ Not knowledgeable at all
- ☐ Slightly knowledgeable
- ☐ Moderately knowledgeable
- ☐ Very knowledgeable
- ☐ Extremely knowledgeable

4. How much do you understand about the importance of an oral care protocol in the prevention of hospital acquired pneumonia?

- ☐ Not knowledgeable at all
- ☐ Slightly knowledgeable
- ☐ Moderately knowledgeable
- ☐ Very knowledgeable
- ☐ Extremely knowledgeable

5. What is the typical duration when you provide oral care using a toothbrush or with the toothbrush kit?

- ☐ 1 minute
- ☐ 2 minutes
- ☐ Do not pay attention with the time
- ☐ _____

6. Do you perform an oral care assessment prior to providing oral care and document in the electronic medical record?

- ☐ Yes
- ☐ No
- ☐ _____

7. How often do you assess the oral cavity and document in the electronic medical record?

- ☐ Every 4 hours
- ☐ Four times a day
- ☐ Twice a shift
- ☐ Once every shift

8. Given all the nursing task to complete for the care of the acute stroke patient, please rate the priority level of care. Choose one.

- ☐ High priority
- ☐ Moderate priority
- ☐ Low priority

10. What items of the oral cavity do you assess? Check all that apply

- ☐ Redness
- ☐ Swelling
- ☐ Dry Mouth
- ☐ Oral mucosa tear, ulcerations, abrasions, and cracks
- ☐ Lack of saliva
- ☐ Dental carries
- ☐ Bleeding
- ☐ Tenderness
- ☐ Other _____

12. Are there any barriers you can identify in providing oral care to the non-ventilated acute stroke patients?

Check all that apply

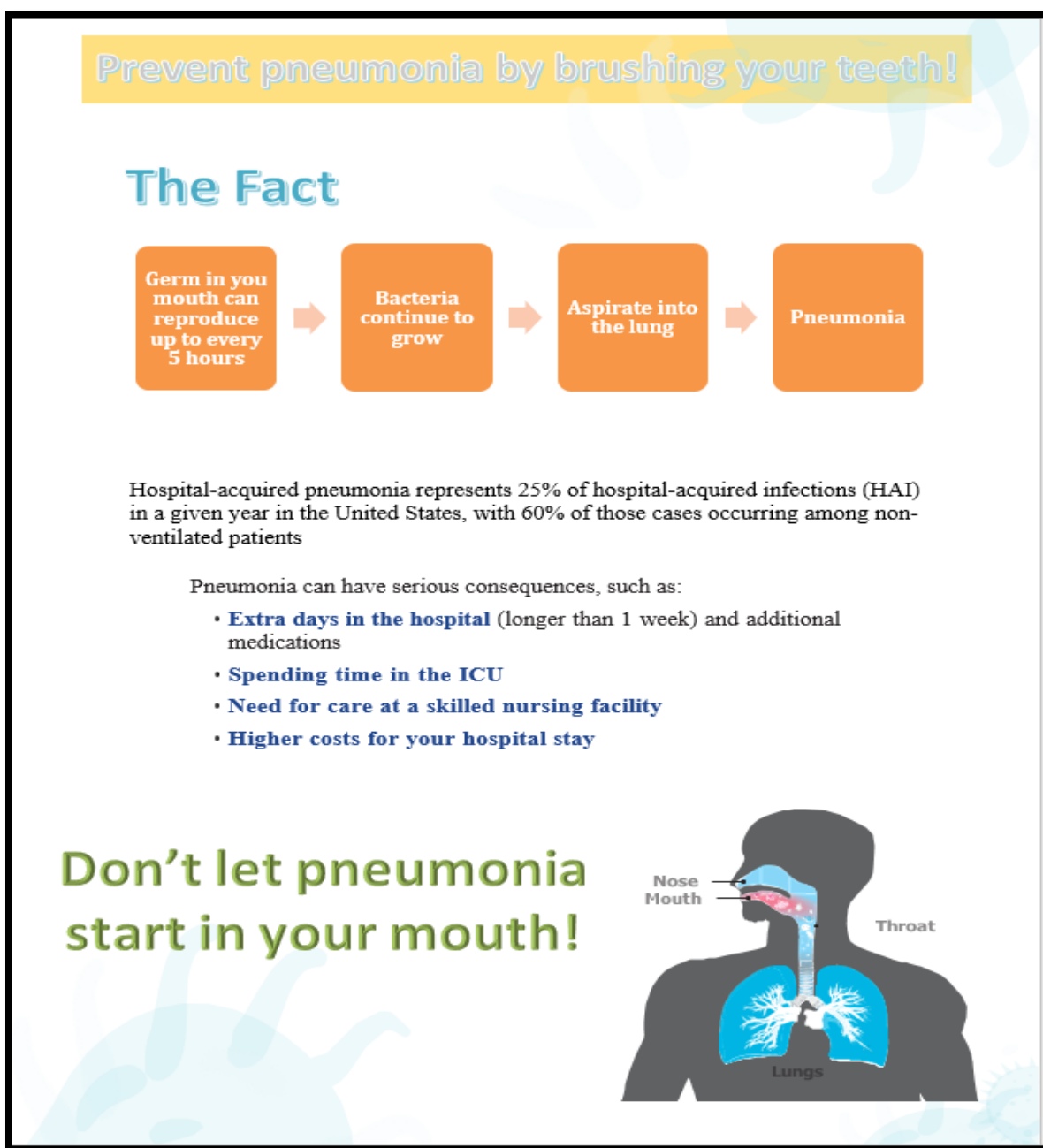
- ☐ Time constraints in performing oral care

- ☐ Inadequate staff resources
- ☐ Inadequate supplies to perform oral care
- ☐ Inadequate training to perform oral care
- ☐ Other _____

Appendix J

Oral Care Patient Education Handout

Figure 5

Oral Care Protocol Audit Tool Page 1 of 2

Appendix K

Oral Care Patient Education Handout

Figure 6

Patient Education Tool Page 2 of 2

Prevent pneumonia by brushing your teeth!



A proven way you and your nurse can partner to prevent pneumonia is **TAKING CARE OF YOUR MOUTH**



1. BRUSH your teeth for 1-2 minutes **after meals** and **at bedtime**. Don't forget to brush your tongue!

Brushing removes sticky film in your mouth that contains bacteria.



2. RINSE with a mouthwash (30 seconds, as directed)

Rinsing kills bacteria and helps clean parts of the mouth that brushing can't reach.



3. USE a lip moisturizer

Using a non-petroleum-based lip moisturizer prevents cracked lips.

Brushing your teeth 4 times a day after each meals & before bedtime along with using a rinse 2 times a day* helps keep the overall germ load in your mouth low, which lowers the risk of acquiring pneumonia, and may help you avoid a longer stay in the hospital

Appendix L

List of ICD 10 Codes for NV-HAP

Table 1

ICD 10 Codes for NV-HAP

ICD-10 Code	Definition
J189	Pneumonia, unspecified
J151	Pneumonia due to Pseudomonas
J15211	Methicillin susceptible pneumonia due to Staphylococcus aureus
J15212	Methicillin resistant pneumonia due to Staphylococcus aureus
J17	Pneumonia in diseases classified elsewhere
J14	Pneumonia due to Hemophilus influenzae
A403	Pneumonia due to Streptococcus pneumoniae
J156	Pneumonia due to other aerobic Gram-negative bacteria
J1289	Other viral pneumonia
B440	Invasive pulmonary aspergillosis
J15.0	Pneumonia due to Klebsiella pneumoniae
J15.5	Pneumonia due to Escherichia coli
J168	Pneumonia due to other specified infectious organisms
J154	Pneumonia due to other streptococci
J1282	Pneumonia due to coronavirus disease 2019
J159	Unspecified bacterial pneumonia
J690	Pneumonitis due to inhalation of food and vomit
J698	Pneumonitis due to inhalation of other solids and liquids
J181	Lobar Pneumonia, unspecified organism
J188	Other pneumonia, unspecified organism

Appendix M

Table 2: Nurses' Knowledge of the Importance of an Oral Care Protocol to Prevent Non-Ventilator Hospital-Acquired Pneumonia (NV-HAP)

Table 2

Knowledge of an Oral Care in the Prevention of NV-HAP

Category	Pre-Implementation	Post -Implementation
Not Knowledgeable	2.50%	0.00%
Slightly Knowledgeable	12.5%	9.38%
Moderately Knowledgeable	42.5%	46.88%
Very Knowledgeable	32.5%	43.75%
Extremely Knowledgeable	10.00%	0.00%

Note. The nurses' survey responses on knowledge of an oral care protocol in preventing NV-HAP were higher in post-implementation with a moderate to very knowledgeable category of responses.

Appendix N

Table 3 Nurses' Survey Responses on the Duration of Oral Care

Table 3

Duration of Oral Care

Observed	Pre-Implementation	Post -Implementation
1 Minute	12.50%	15.15%
2 Minutes	37.50%	51.52%
Paid no Attention	50.00%	33.33%

Note. Survey responses of nurses showed an increase in the duration of oral care by 2 minutes at post-implementation compared to baseline when nurses paid no attention to the duration of oral care.

Appendix O

Table 4 Nurses' Survey Responses on the Prioritization of Oral Care

Table 4

Prioritization of Oral Care

Observed	Pre-Implementation	Post -Implementation
High	22.5%	27.2%
Moderate	32.5%	48.48%
Low	45.00%	24.24%

Note. Survey responses by the nurses during post-implementation showed an increased value in the prioritization of oral care to moderate from low at baseline.